

Department of Mathematics and Statistics, Queen's University

MTHE 351

Probability I - Fall 2026

Instructor: Fady Alajaji
E-mail: fa@queensu.ca
TA: Ananya Omanwar (22asol@queensu.ca)
Markers: Ananya Omanwar (22asol@queensu.ca) and Jonas Schuppert (26qtd1@queensu.ca)
Webpage: [OnQ](#)
Textbook: F. Ghahramani *Fundamentals of Probability with Stochastic Processes, Fifth Edition*, Chapman and Hall/CRC, 2024 (Campus bookstore [link](#))(~ US\$170)
Lectures: Tuesday 8:30 (Dupuis 215), Wednesday 12:30 (Jeff 126), Friday 9:30 (Dupuis 215)
Tutorial: Wednesday 1:30 (Jeff 126)
Office Hours: Wednesday 10:30 - 11:30
Course units: 3.5 units
Grading: *Homeworks:* 10% – *Quiz 1:* 20% – *Quiz 2:* 20% – *Final Exam:* 50%

- There will be a total of 10 **Homework** assignments to be submitted via **Crowdmark**.¹
- **Quiz 1** is scheduled in Week 5 for **Friday, October 9, 9:30 - 10:20**. **Quiz 2** is scheduled in Week 8 for **Wednesday, November 4, 1:30 - 2:20**. There will be no makeups for the quizzes. If a student misses a quiz due to severe illness, a personal tragedy or other valid reasons, then its weight will be move to the final exam.
- The **Final Exam** will be run by the Exams Office.
- [Queens Official Grade Conversion Scale](#).

Course Outline:

- *Basic Concepts of Probability Theory:* axioms of probability; counting; conditional probability; law of total probability and Bayes' rule; independence of events (Chapter 1, Sections 2.1-2.4, Chapter 3 of text: Weeks 1-4).
- *Discrete Random Variables:* random variables; distribution functions; expectation and variance of a discrete random variable; uniform, Bernoulli, binomial, negative binomial, Poisson, and geometric distributions (Chapters 4 and 6 of text: Weeks 4-8).
- *Continuous Random Variables:* probability density functions; functions of random variables; expectation and variance of a continuous random variable; uniform, normal, and exponential random variables (Chapter 5 and Sections 7.1, 7.2 and 7.4 of text: Weeks 8-10).
- *Pairs of Random Variables:* joint distributions; independent random variables; conditional distribution and expectation; functions of two random variables (Sections 8.1, 8.4 and 8.5 of text: Weeks 10-12).

¹ Turnitin will not be used in this course.

Special Accommodations: If you require special accommodations or special consideration, please submit your forms to [an Advisor](#) at the [the Smith Engineering Academic Accommodation and Consideration Office](#).

Academic Integrity Policy: Academic integrity is constituted by the five core fundamental values of honesty, trust, fairness, respect and responsibility. These values are central to the building, nurturing and sustaining of an academic community in which all members of the community will thrive. Adherence to the values expressed through academic integrity forms a foundation for the “freedom of inquiry and exchange of ideas” essential to the intellectual life of the University. Students are responsible for familiarizing themselves with the regulations concerning academic integrity and for ensuring that their assignments conform to the principles of academic integrity. Information on academic integrity is available on this [link](#).

Departures from academic integrity include plagiarism, use of unauthorized materials, facilitation, forgery and falsification, and are antithetical to the development of an academic community at Queen’s. Given the seriousness of these matters, actions which contravene the regulation on academic integrity carry sanctions that can range from a warning or the loss of grades on an assignment to the failure of a course to a requirement to withdraw from the university.

Use of Generative Artificial Intelligence (GenAI) Tools: The use of GenAI tools is *permitted only for homework assignments* and only as an educational tool to support and enhance learning. Students may use GenAI to clarify concepts, brainstorm ideas, or receive feedback on their work. However, they are expected to *critically evaluate any AI-generated content and ensure that all submitted work reflects their own understanding and effort*.

Calendar Description:

Introduction to probability theory and its applications in engineering science: basic concepts of probability, counting, conditional probability, Bayes’ rule, independence; probability models; random variables, distribution functions, probability mass functions, probability density functions; expectation, variance, moments; jointly distributed random variables; transformations of random variables. Distributions: Bernoulli, binomial, geometric, negative binomial, Poisson, uniform, exponential, normal. Applications: elementary stochastic processes, noisy communication channels. (Lec: 3, Lab: 0, Tut: 0.5).

Course Learning Outcomes (CLOs)

CLO	Description	Indicator
CLO 1	Computing the probability density function of a function of a random variable	KB-Mathematics (Computational proficiency)
CLO 2	Computing probabilities of events defined by random variables and computing moments of random variables	KB-Mathematics (Computational proficiency)
CLO 3	Computing joint probability density functions of random variables	KB-Mathematics (Computational proficiency)
CLO 4	Reasoning probabilistically in order to solve problems of sampling with or without replacement	KB-ES
CLO 5	Reasoning probabilistically in order to solve problems of quality control	KB-ES
CLO 6	Rigorously establishing relations between probabilities or conditional probabilities of various events	KB-Mathematics (Mathematical rigour)